

Roll No.

TPE-301

B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
THERMODYNAMICS
Time : Three Hours
Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Discuss first and second laws of thermodynamics with the expressions ?
Draw the P-V-T behavior of pure fluids and discuss the terms saturated phase and superheated phase from the diagram. (CO1, CO2)
(b) Define the following : (CO1, CO2)
(i) Isochoric process
(ii) Intensive and Extensive Properties with examples
(iii) Degree of freedom
(iv) Enthalpy and Internal energy
(v) Homogeneous system
(vi) Entropy of the system

P. T. O.

- (c) What are the two propositions involved in the Carnot principle ? Discuss Ideal gas as the Carnot engine working substance. (CO3)
2. (a) Illustrate Adiabatic process and hence derive an expression for the work done under adiabatically condition of the system. Specify the symbol used. (CO3, CO4)
- (b) Define polytropic process. What is the process called when value of $n=1$ and $n=\infty$? Hence derive the equation for work done, internal energy and heat supplied. (CO1, CO4)
- (c) What do you mean by COP of a refrigerator ? Establish relationship between COP of refrigerator and efficiency of a heat engine.

Solve :

It is required to freeze 1 kg water at 273 K by means of a refrigeration machine which operate in the surroundings 300 K. The latent heat of fusion of ice at 273 K is 884. 11 kJ/kg.

Determine : (CO1, CO5)

- (i) The minimum amount of work required
- (ii) The heat given up to the surroundings.
3. (a) Classify the thermodynamics properties of pure fluids ? How are Maxwell's equations helpful ? Derive the Maxwell's equations from the exact differential equations of the four energy properties. (CO1)
- (b) Establish the entropy-heat capacity relationships and hence determine the increase in entropy of solid magnesium when the temperature is increased from 300 K to 800 K at atmospheric pressure. The heat capacity is given by the equation as : (CO1, CO2)

$$C_p = 26.04 + 5.586 \times 10^{-3} T + 28.476 \times 10^{-4} T^{-2}.$$

(3)

- (c) Illustrate the classification of thermodynamic properties. Discuss the energy properties in terms of entropy of the system. (CO2, CO5)
4. (a) Explain Gibbs free energy and hence derive the modified equations for Internal energy and enthalpy by using the energy and Maxwell's relations. Specify the symbol used. (CO1, CO5)
- (b) Define the inversion point of Joule-Thompson co-efficient (μ). Prove that the Joule-Thompson co-efficient (μ) in Throttling Process can be expressed by the expression as : (CO4, CO5)

$$\mu = \frac{\left[T \left(\frac{dV}{dT} \right)_P - V \right]_{\text{constant } h}}{C_p}$$

- (c) Explain fugacity and fugacity coefficient ? Derive the expression for the determination of fugacity of pure gases by using compressibility factor Z. (CO1, CO6)
5. (a) Derive an expression for the fugacity coefficient of a gas obeying the equation of state $P(V - b) = RT$ and estimate the fugacity of ammonia at 10 bar and 298 K, given that $b = 3.703 \times 10^{-5} \text{ m}^3/\text{mol}$. (CO4, CO6)
- (b) Discuss "The Linde process" for gas-liquefaction along with the component construction arrangements (diagram) of equipments used for the process. (CO4, CO5)
- (c) What do you mean by refrigerant ? To design the refrigeration process the types of refrigerant is very important. What are the characteristic parameters on which the choice of refrigerant for the refrigeration process is to be considered ? Discuss in detail. (CO2, CO6)

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21/12/23
1:30 to 4:30

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TPE-302

**B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

APPLIED GEOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the different types of sand dunes. (CO1)
(b) What is river meandering ? How the ox-bow lake is formed ? Explain with the help of a diagram. (CO1)
(c) List and describe the any *four* landforms developed by the glacier erosion. (CO1)
2. (a) Explain the exogenic and endogenic processes. (CO1)
(b) What are different types of plate boundary ? Explain with the help of diagram. (CO1)

P. T. O.

(2)

- (c) What is difference between mineral and glass ? Describe the *four* physical properties of mineral for identification of minerals as handspecimen. (CO2)
3. (a) What is the process of formation of Igneous rock ? What is the difference between intrusive and extrusive igneous rock ? Write the name of *two* intrusive and *two* extrusive igneous rocks. (CO3)
- (b) What is a sedimentary rock ? Describe the process of formation of sedimentary rock. (CO3)
- (c) What is unconformity ? Explain different types of unconformity.
4. (a) List and describe the laws of stratigraphy ? (CO4)
- (b) Discuss the significance of geological structures in petroleum industry. (CO6)
- (c) Draw the geological time scale showing different era, periods and epochs. (CO4)
5. (a) What is fold ? Describe hinge line, inflexion point, limb and fold axis. Differentiate between anticline and syncline fold. (CO5)
- (b) Give the classification of fold on the basis of interlimb angle and fold axis. (CO5)
- (c) What is difference between fault and fracture ? What is normal fault and reverse fault ? Explain with diagram. (CO5)

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TPE-303

**B. TECH. (PE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

FLUID AND PARTICLE MECHANICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Show that the equation $v = u + at$ is dimensionally homogeneous.

(CO1)

(b) Derive Bernoulli equation by integrating Euler equation of motion.

(CO1)

(c) Explain dimensional homogeneity ? Where is it applied ?

(CO1)

2. (a) What is head ? What are the three heads in Bernoulli's equation ?

(CO1, CO2)

(b) Show that the shear stress in a steady laminar flow in a circular pipe is linear.

(CO1, CO2)

(c) What are major and minor losses ? How can major loss be estimated ?

(CO1, CO2)

P. T. O.

(2)

3. (a) What are Pi terms in Buckingham Pi theorem ? How do you choose the variables for the Pi terms ? (CO2, CO3)
(b) Explain how dimensional analysis can be beneficial for comparing the performance of scale model with the prototype. (CO2, CO3)
(c) What is Reynolds number ? How does the flow characteristics change with Reynolds number ? Explain with diagram. (CO2, CO3)
4. (a) On what principal does the centrifugal pump works ? What are its basic parts ? (CO3, CO4)
(b) Why is reciprocating pump called positive displacement ? What are its advantages and disadvantages ? (CO3, CO4)
(c) Draw a diagram and explain the working of a double acting reciprocating pump. (CO3, CO4)
5. (a) Explain fluidization with diagram and indicate its use. (CO5, CO6)
(b) What are the different types of crushers ? Why are they used ? (CO5, CO6)
(c) What is screening ? Explain effectiveness of a screen. (CO5, CO6)

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TPE-304

**B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

GEOMATIC ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Surveying and explain the basic principle of surveying with the help of neat diagram. (CO1)
- (b) The area of the plan of an old survey plotted to a scale of 10 meters of 1 cm measures now as 100.2 sq. cm as found by a planimeter. The plan is found to have shrunk so that a line originally 10 cm long now measures 9.7 cm only.
Find : (CO1)
 - (i) Shrunk scale
 - (ii) True area of the survey
- (c) Explain difference between Prismatic compass and Surveyor's compass with a sketch. (CO1)

P. T. O.

(2)

2. (a) The fore bearings of lines AB, BC, CD and DE, are $45^{\circ}30'$, $120^{\circ} 15'$, $200^{\circ}30'$ and $280^{\circ} 45'$. Find angles B, C and D. (CO2)
(b) Enlist the temporary adjustments to be carried out for theodolite. Explain three screw head method of levelling. (CO2)
(c) Explain any two of the following terms related to chain surveying with the help of neat diagram : (CO2)
Line ranger, Cross staff, Optical Square, Clinometer.
3. (a) Explain plane table survey and various instruments used in plane table survey. (CO3)
(b) Enlist the various sources of errors in a traverse survey. (CO3)
(c) Write a short note on Traverse survey. What are the various types of traverse survey ? (CO3)
4. (a) Write a short note on techniques of photo interpretation. (CO4)
(b) Explain the terms SCALE, TILT and height displacement related to aerial photography. (CO4)
(c) Write in brief about the use of stereoscope and parallax bar. (CO4)
5. (a) Explain the principle behind the remote sensing and various applications of remote sensing. (CO5)
(b) Explain the terms GIS and remote sensing. Write in brief about scope of GIS and remote sensing in mineral exploration. (CO5)
(c) Write in brief about spatial and non-spatial data and digital image processing. (CO5)

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TMA-307

B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
ADVANCED MATHEMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the Z-transform. Find the Z-transform of a discrete unit step sequence : (CO1)

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

- (b) Find the Z-transform of $\{f(n)\}$, where : (CO1)

$$f(n) = \begin{cases} 2^n & \text{for } n \geq 0 \\ 3^n & \text{for } n < 0 \end{cases}$$

P. T. O.

- (c) Using Partial fraction method find the inverse Z-transform of the given function : (CO1)

$$F(z) = \frac{z}{(z-1)(z-2)}$$

2. (a) Find all values of k such that the function : (CO2)

$$f(z) = e^x [\cos ky + i \sin ky]$$

is analytic.

- (b) Check the real part of : (CO2)

$$f(z) = u(x, y) + iv(x, y)$$

where $u(x, y) = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. If yes, then find the analytic function $f(z)$.

- (c) Define the following terms : (CO2)

- (i) Cauchy-Riemann equations in Cartesian coordinates
- (ii) Analytic function
- (iii) Harmonic function
- (iv) Complex integration

3. (a) Using Cauchy's integral formula to compute the value of the integral :

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$$

where $C : |z| = 3$. (CO3)

- (b) To evaluate the value of the integral $\oint \frac{e^z}{z^2 - 1} dz$, along the closed curve

$C : |z| = 2$. (CO3)

(c) Show that the function :

$$f(z) = u + iv$$

where

$$f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is not analytic at $z = 0$, although Cauchy-Riemann equations are satisfied at $z = 0$. (CO2)

4. (a) Define the following : (CO4)

- (i) Random Variable
- (ii) Probability mass function
- (iii) Probability density function

(b) If the probability density function of a continuous random variable X is given by : (CO4)

$$f(x) = \begin{cases} 3x^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the mean, variance, and standard deviation.

(c) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line of y on x and x on y . (CO4)

Time periods (x)	Diastolic blood pressure (y)
0	72
5	67
10	70
15	65
20	66

5. (a) The first four moments about the mean of a frequency distribution are 0, 7.058, 36.151, and 408.738. Calculate β_1 , β_2 , and comment upon the Skewness and Kurtosis of the distribution. (CO5)
- (b) The number of airplane landings at an airport in a minute (X) and their probabilities are given in the following table : (CO5)

$X = x_i$	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Calculate the mean and variance of the random variable X .

- (c) Define the Kurtosis, Skewness, Correlation and Regression Analysis.

(CO5)

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TPE-311

**B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
INTRODUCTION TO PETROLEUM OPERATIONS**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is petroleum exploration ? List the different types of petroleum exploration. Describe the operations involved in seismic methods of exploration. (CO1)
- (b) What are the salient feature of well programme ? Discuss. (CO2)
- (c) What is an economic evaluation of an exploratory project ? Discuss. (CO1)
2. (a) What is kick off point in directional drilling ? Describe the three types of well profile in directional drilling with the help of diagram. (CO1)

P. T. O.

(2)

- (b) What is the components of well control system ? What is the difference between kick and blowout ? (CO3)
- (c) What is sticking of pipe ? Explain pack of bridging. (CO3)
- 3. (a) What is well logging and its purpose ? What is open hole and cased hole logging ? In what condition open hole logging is carried out ? (CO4)
- (b) What is drill time, caliper and sonic log ? (CO4)
- (c) Define porosity. What is the difference between absolute and effective porosity ? Which logs are used for porosity determination ? (CO4)
- 4. (a) List and describe the perforation techniques used in oil well. (CO5)
- (b) Describe the artificial methods used in production operations of oil well. (CO5)
- (c) What is a well head assembly. What does a well head do ? What are the component of well head assembly ? (CO5)
- 5. (a) What are different types of drilling fluid ? What are the functions of drilling fluids ? (CO3)
- (b) Describe the operations involved in distillation of crude oil. (CO6)
- (c) Discuss the role of IT in oil and gas industry. (CO6)